

QS Cost — System of Record

See cost trouble **one reporting period early** — while the money is still unspent.

● GREEN



● AMBER

we call the tip *before* it happens

ASP.NET Core 8

React + Vite

Postgres · RLS

Claude copilot

Tower X · 173 cost centres · AED

TEAM

HA

Hazem Ashraf

N

Nassim

AA

Abd El Aziz

AK

Ahmed Khafagy

M

Moayad

— THE PROBLEM

A QS spots the overrun weeks **too late**.

Working in spreadsheets, the problem surfaces at month-end close — **after the invoices are paid and the money is gone.**

- Every cost centre has a planned budget vs earned value vs actual spend.
- Drift hides in the noise until a period closes.
- By then it's a report, not a decision.

THE CONCRETE POUR

~~50,000~~ → **65,000**

A concrete pour budgeted at 50,000 AED that quietly became 65,000 should have been flagged **the moment it started drifting** — not at month-end.

— OUR INSIGHT

Flag it **one period before it tips.**

We don't wait for a cost centre to go AMBER. We rank the still-GREEN centres **most likely to tip next period** — while the budget is unspent and the QS can still act.

• GREEN at period p

on budget, today



we predict the tip

• AMBER at p + 1

CPI breaches 0.95

One honest, verifiable event — not a vague "risk." On the Tower X history, **AMBER $\equiv$ CPI < 0.95** on every live row (0 mismatch): the target is a real threshold crossing, so the prediction is gradeable.

— THE PRODUCT

The spreadsheet, replaced.

A live **system of record** for EVM — computed from Postgres, multi-project, always current.

- 173 cost centres with live BAC · Plan% · Actual% · EV · AC · **CPI** · EAC.
- GREEN / AMBER status the moment a number changes.
- Data entry writes to Postgres and EVM recomputes — no month-end batch.

localhost:5173 · Cost Centres

QS Cost — System of Record

scorer RuleRiskScore@v1 · 173 centres loaded

Live EVM from Postgres · multi-project · data entry replaces the spreadsheet.

Project Tower X (lower-x) (AED) Period Period 12 — forecast cumulative snapshot · active estimate v2

EVM Overview Cost Centres Monthly Capture Periods & Estimate Watchlist Model & Copilot

173 cost centres · period 12 filter BCC...

COST CENTRE	DISCIPLINE	STATUS	BAC	PLAN%	ACT%	EV	AC	CPI	EAC
BCC-ARC-MAS-301	Architectural Fini	AMBER	3,444,671	39.0	30.0	1,033,401	1,090,548	0.948	3,635,161
BCC-ARC-MAS-302	Architectural Fini	GREEN	1,543,233	44.0	35.0	540,132	562,362	0.960	1,606,747
BCC-ARC-MAS-303	Architectural Fini	GREEN	754,585	43.0	43.0	324,472	335,961	0.966	781,304
BCC-ARC-MAS-304	Architectural Fini	AMBER	836,479	45.0	33.0	276,038	291,031	0.948	881,912
BCC-ARC-PAINT-317	Architectural Fini	GREEN	851,359	39.0	39.0	332,030	343,065	0.968	879,654
BCC-ARC-PAINT-318	Architectural Fini	AMBER	333,986	44.0	42.0	140,274	149,635	0.937	356,274
BCC-ARC-PLST-305	Architectural Fini	AMBER	3,037,545	44.0	34.0	1,032,765	1,088,920	0.948	3,202,707
BCC-ARC-PLST-306	Architectural Fini	AMBER	862,443	43.0	38.0	327,728	349,809	0.937	920,551
BCC-ARC-PLST-307	Architectural Fini	GREEN	2,868,274	42.0	37.0	1,061,261	1,100,324	0.964	2,973,850
BCC-ARC-SCRZ-311	Architectural Fini	GREEN	1,870,094	41.0	30.0	561,028	578,414	0.970	1,928,047
BCC-ARC-SCRZ-312	Architectural Fini	GREEN	555,955	45.0	45.0	250,135	251,821	0.993	559,602

A ranked triage list.

"Which GREEN cost centres are about to tip to AMBER?"

- Scored by a **frozen, transparent rule** — `RuleRiskScore@v1`, no black box.
- $0.7 \cdot \text{spend-vs-progress gap} + 0.3 \cdot \text{CPI-proximity}$
- Click any row → the **cost-centre drawer** opens: variance + AI correction actions.

TOP GREEN CENTRES AT RISK · PERIOD 12

CENTRE	DISCIPLINE	CPI	RISK
BCC-IND-QC-1805	Indirect/Prelims	0.950	+3.29
BCC-IND-SAF-1804	Indirect/Prelims	0.951	+3.22
BCC-ARC-PLST-305	Architectural	0.952	+2.87
BCC-ARC-MAS-302	Architectural	0.960	+2.41
BCC-STR-RC-118	Structural	0.958	+2.18

Risk = spending points ahead of progress, weighted toward centres already near the 0.95 line.

We let the app **grade itself**.

The Proof tab rewinds to an earlier period, shows the model **only the past**, lets it flag the top-5 GREEN centres — then reveals what actually happened next month.

Rewind

to period 5 — the model sees only history up to here, no leakage.

Flag

the 5 GREEN centres most likely to tip AMBER next period.

Reveal

step forward one month → the app scores its own calls, hit by hit.

"Caught a full reporting period early — **before the invoices landed**."

— VALIDATION · HONESTY

Validated, not asserted.

OUR RULE

45%

precision@5 — of the 5 we told the QS to chase, how many really tipped

vs

BEST CPI BASELINE

35%

the strongest CPI-native heuristic we could build

8 rolling-origin folds

117 real GREEN→AMBER transitions

74 centres · 12 periods

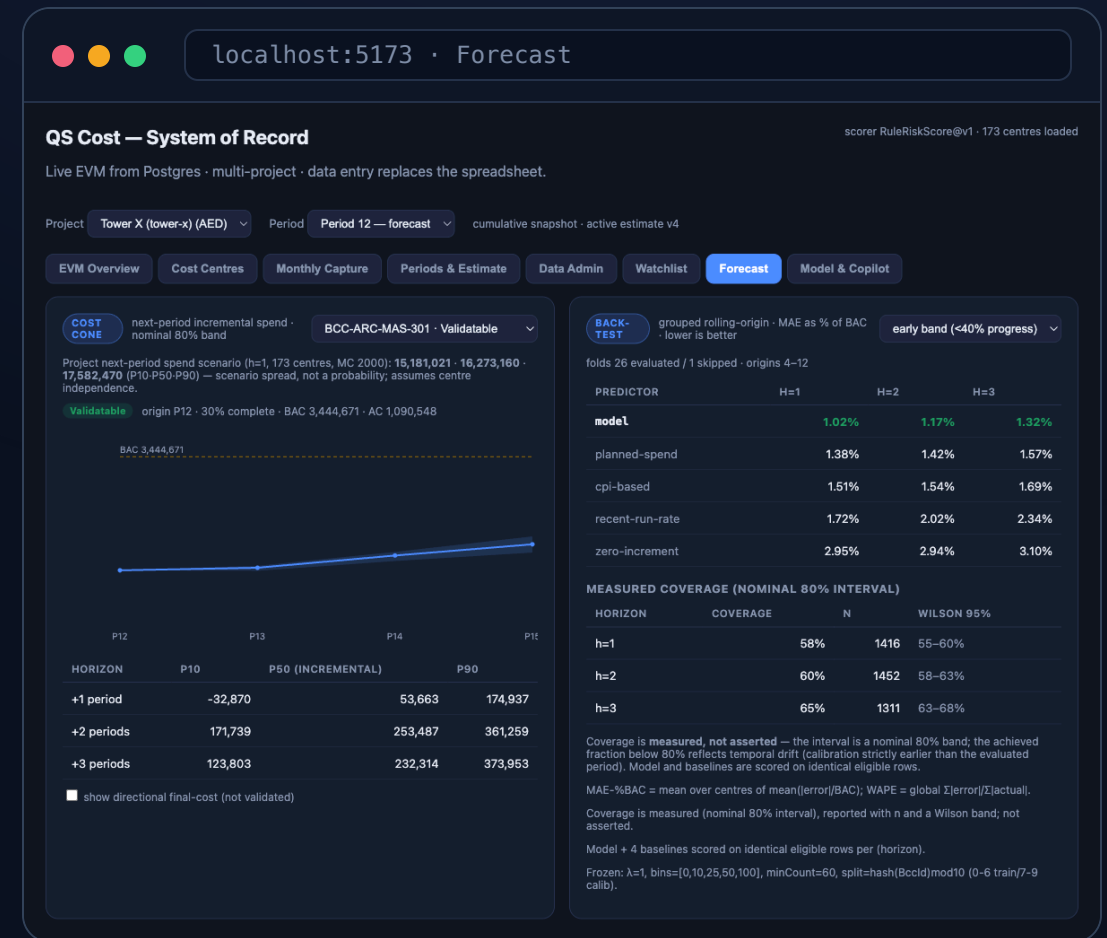
Leakage-safe — each period scored by a model trained only on its past

The rule is **frozen and predeclared**, so nothing is tuned on the eval folds — the back-test is unbiased. And we **rejected the naive final-cost EAC**: this project has almost no finished work, so a final-cost number couldn't be honestly validated. Single project — Tower X's number, not a universal benchmark.

Next-period spend, with a real band.

"How much will we spend next period — and what's the range?"

- Ridge P50 forecast + a **split-conformal P10–P90 band**.
- Coverage is **measured, not assumed** — reported with the achieved fraction.
- Beats **4 baselines** (planned-spend, CPI-based, run-rate, zero) on MAE-%BAC.

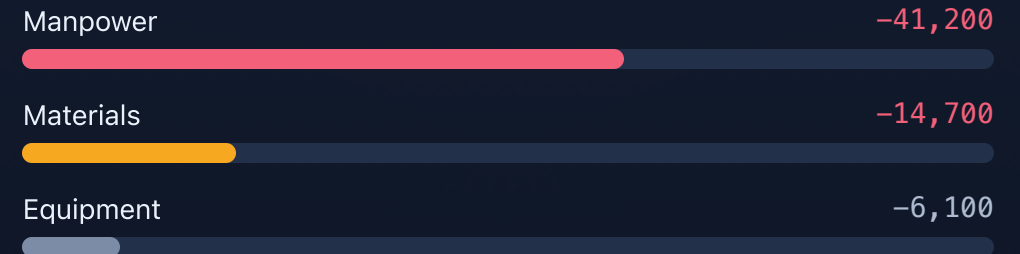


Where is the over-run coming from?

"Which resource is this package's variance attributed to?"

- Decomposes CV into **resource-category lanes** + a schedule (SV) lane.
- Ties out additively: $\sum CV_r + \text{residual} == CV$.
- The *cause* is labelled a **hypothesis** — the data can't split price from productivity.

CV ATTRIBUTION · BCC-ARC-MAS-301



Schedule lane: **SV negative** — behind plan. Dominant contributor:

Manpower assumption-based

— 4

ESTIMATE ASSUMPTION STRESS TEST

Catch trouble **before** award.

The angle others miss — attack cost risk **before a single dirham is spent**, by reviewing the estimate's own assumptions.

CLASS 1

Reconciliation tie-out

Does the resource-level estimate roll up to the BOQ contract total? A hard cross-check.

CLASS 2

Assumption flags

Cohort-gated: aggressive Output Norm, thin unit rate, thin contingency vs peers.

CLASS 3

Peer benchmark

Retrospective only — where this package sits against comparable ones. Honest scope.

Not an "objective under-pricing" oracle — one project can't prove that. It surfaces assumptions **for a QS to review**, with the evidence attached.

Ask it in plain English.

"Tools compute, the model narrates."

- 12 **read-only tools** do every calculation — the model never does the AED math. One of them is the **unit-rate what-if**. →
- On an AMBER centre it **drafts prioritised correction actions** + a 3-month **what-if savings band** — the AI supplies only the reduction %, the engine does the AED.
- Ask why **any** centre drifted — an already-AMBER one returns its **trajectory**: when it crossed 0.95, how long it's held, which way CPI moves.
- A **Sources panel** cites every tool call — resolved period/filter, rows excluded, source row IDs.
- Claude Sonnet 5 · Microsoft Agent Framework; **auth-before-LLM**, refuses off-topic — no hallucinated money.

QS Copilot

QS Cost Copilot

Ask about the watchlist in plain English. Answers are grounded in the same read-only tools.

Which centres are about to tip to AMBER in period 12, and why?

Here are the top GREEN cost centres most at risk of tipping AMBER in Period 12 (live forecast), ranked by risk score:

Rank	Centre	Discipline	Risk Score
CPI	Gap (pp)	Driver	
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1	**BCC-IND-QC-1805**		
Indirect/Prelims	0.999	0.950	+3.29
Spending 3.3pp ahead of progress; CPI sitting right on the 0.95 line and still falling			
2	**BCC-IND-SAF-1804**		
Indirect/Prelims	0.998	0.951	+3.22
Spending 3.2pp ahead of progress; CPI just above the 0.95 threshold			

Ask about a centre or the watchlist

Ask

"What if we renegotiate?"

"If we get the subcontractor to 299 AED/unit from next period — where does this centre land?"

IN The assumption

The QS's rate — **not a figure from the sheets**. Echoed back with its effective period, so the assumption is auditable.

ENGINE Reprice remaining

Remaining quantity repriced at that rate, **holding recent physical pace** — a price renegotiation changes cost, not tempo.

OUT The scenario

3-period spend path + cost-to-complete, **scenario final cost and VAC**. All arithmetic in C#; the model only narrates.

THE CONTRAST IS THE POINT

planned

BAC ÷ budget qty

realized today

AC ÷ earned qty

assumed

the QS's number

scenario, not a forecast

Every figure is a direct arithmetic consequence of the stated assumption — never presented as the validated forecast. For a data-driven number, the copilot routes to the conformal forecaster instead.

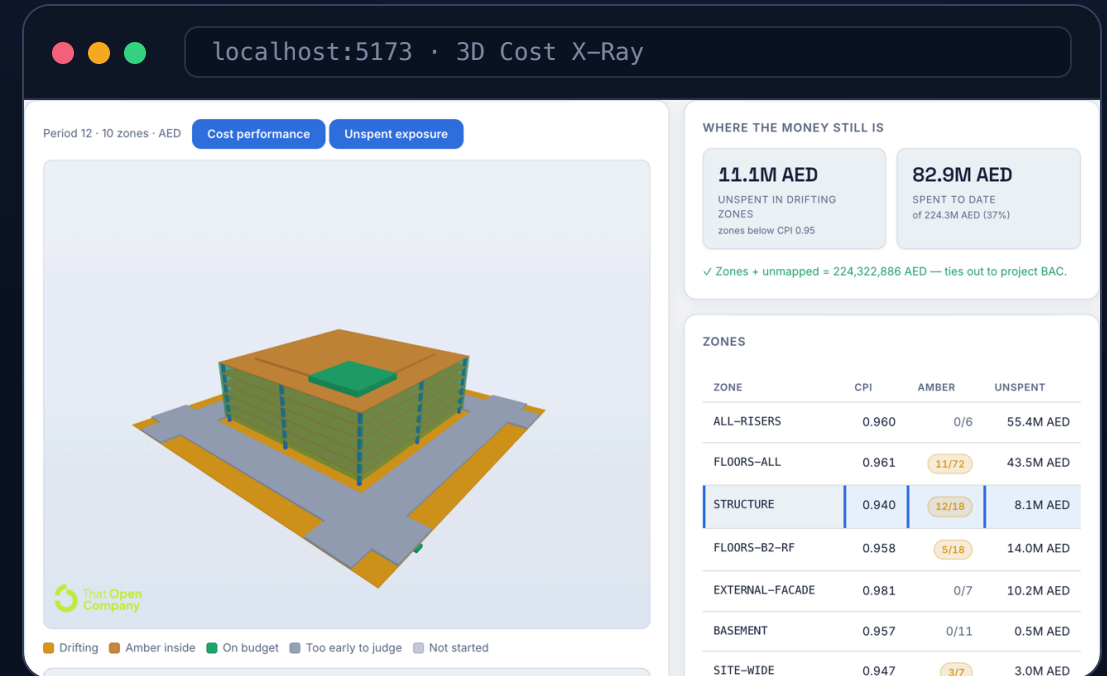
The watchlist, on the building.

"Which centre is drifting?" was never the hard question. "Where — and is there still money to save?" is.

- 19.96M AED sits in AMBER centres and is **not yet spent**.
- STRUCTURE is the damage — CPI 0.940, 78% spent, **only 8.1M** left to save.
- FLOORS-ALL reads **GREEN** at 0.961 — while holding **11 AMBER** centres and 43.5M unspent.

aggregation hides trouble

A healthy-looking zone can contain eleven unhealthy centres — which is the argument for the click-through into the existing drawer. Tower X has no published model, so the massing is generated from the BOQ: every dimension traces to a priced line, and the derivation is on screen.



Point it at **any** model. Get a price.

*Cost history belongs to one building. A **rate library** travels to every building you can measure.*

- 4.64M AED priceable at Tower X's rates — on a building nobody here has ever priced.
- 58% of elements measurable. The other 42% is **shown, not hidden**.
- Zero** IfcElementQuantity in a real Revit export — the textbook take-off returns nothing.

real BIM is messy

The quantities exist only inside the exporter's own parameter groups — **in Spanish**. 375 beams measured but unpriceable (this BOQ prices no beam concrete); 619 rebar carrying no quantity at all. Priced + unpriced + unmeasured ties out to the model's own element count.

The screenshot shows a web application titled "localhost:5173 · IFC Take-off". The main area displays a 3D model of a school building, labeled "school_str.ifc". A button "Load another IFC" is visible. To the right, a summary box shows "PRICED AT TOWER X'S RATES" with a total of "4.6M AED" (PRICEABLE SCOPE: 4,638,842 AED) and "58% MEASURABLE" (883 of 1526 elements). Below this, a green checkmark indicates "508 priced + 375 unpriced + 643 unmeasured = 1526 elements in the model." A table titled "WHAT COULD BE PRICED" lists the following data:

CLASS	QUANTITY	RATE	AMOUNT
IFCSLAB · 2.06	2,735.5 m³	1122.59	3,070,815 AED
IFCSLAB · 2.11	6,761.8 m²	181.80	1,229,315 AED
IFCWALL · 2.05	127 m³	1459.66	185,443 AED
IFCCOLUMN · 2.04	112.3 m³	1364.28	153,268 AED

At the bottom, a note states: "This is not Tower X. It is a school's structural model (Autodesk Revit sample, IFC4) being priced with Tower X's rate library. The two buildings are unrelated — what is being demonstrated is that a rate library travels to..."

We asked the wrong question — and found out in time.

We set out to test: **does physical neighbourhood predict cost drift?** First we checked whether the data could answer it.

IT COULD NOT

8 / 10

zones hold a single trade

0 / 18

trades span more than one zone

Zone_Area is a coarsening of Discipline. A "zone-neighbour" feature measures **trade**, not space.

- So we ran **both** questions, each labelled for what it really measures — with the peer weight **predeclared, not fitted**. A weight tuned on the folds that report the result wins by construction.

a candidate, not a promotion

Without that check we would have published **+12.5pp with the wrong explanation attached**. The headline rests on 40 ranked slots, so RuleRiskScore@v1 stays deployed.

PRECISION@5 · 8 ROLLING-ORIGIN FOLDS

RANKER	P@5	VS RULE
Rule (deployed)	45.0%	—
Best CPI-native baseline	35.0%	-10.0
Peers · same trade	57.5%	+12.5
Peers · same place, different trade	42.5%	-2.5

Trade peers predict drift. Space does not.

The genuinely spatial variant is **identical to the rule on 7 of 8 folds** — it changes almost nothing.

— UNDER THE HOOD

Production-shaped, not a notebook.

API ASP.NET Core 8

Layered Domain / Infrastructure / Core / Web.API — analytics reimplemented in **C#**,
no Python at runtime.

UI React 18 + Vite

Hand-rolled CSS, **zero UI-framework dependencies.** 12 tabs over one coherent product.

DB Postgres · RLS

9 migrations, row-level security, multi-tenant. Each project's data is isolated per user.

AI Claude agent

Microsoft Agent Framework over Claude Sonnet 5. Read-only tools, auth resolved before the LLM runs.

~100 xUnit tests — data-contract, leakage, metrics, copilot eval

Every feature passed an **OpenAI Codex review**

— WHY IT'S CREDIBLE

Honest where it counts.

Validated, not asserted

Rolling-origin back-test on real history — 45% vs 35% baseline, reported per-fold.

Auditable AI

Tools compute + cite every figure; auth-before-LLM; refuses off-topic. No hallucinated money.

Honest about limits

Rejected the un-validatable final-cost EAC; labels directional vs validated; single-project scope stated.

One coherent product

Watchlist, forecast, variance, stress-test and copilot — a system of record, not four disconnected demos.

— THANK YOU

See cost trouble one period early.

A validated early-warning watchlist, a calibrated forecaster, variance attribution, a pre-award stress test, and an auditable copilot that drafts the corrective actions — over one live EVM system of record.

Next: ML.NET challenger · more projects